ZymagenX™

Latiglutenase – Evidenced Safe and Effective Treatment for Celiac Disease

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Contact
Jack A. Syage, PhD
CEO
714 357-0631
jsyage@zymagenx.com

Corporate Office
1600 Dove Street, Suite 330
Newport Beach, CA 92660
www.zymagenx.com

Section I

Company Overview

Highlights

Latiglutenase – Leading Candidate for Treatment of Celiac Disease

1. Completed Phase 2 program shows convincing evidence for safety and symptomatic improvement and histologic protection for celiac disease
2. Leading candidate with first-to-market opportunity addressing an unmet need in a multi-billion-dollar market
3. End-of-Phase 2 Meeting completed and Phase 3 trial in preparation
4. Highly experienced and successful management team and scientific advisory board
5. Significant value creation and early exit potential

Celiac Disease

Genetically predisposed autoimmune disease that afflicts ~1% of the world's population

FACTS ABOUT CELIAC DISEASE

Genetically predisposed autoimmune disease that afflicts ~1% of the world's population

Chronic and debilitating gastrointestinal problems and other long-term health issues

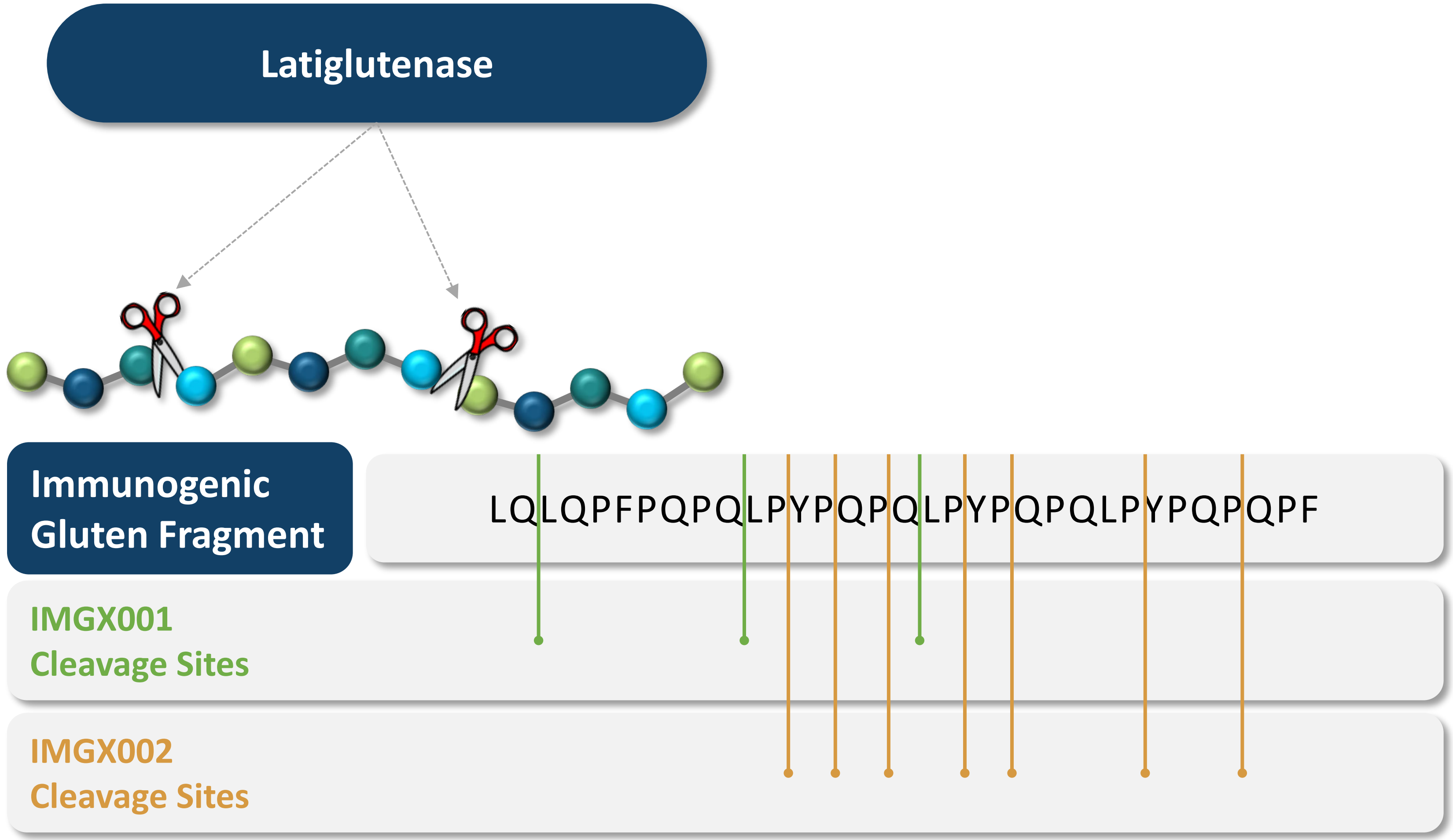
Only current treatment is a gluten-free diet (GFD) which is difficult to implement, costly and often ineffective

Despite adherence to a GFD, CeD patients typically consume 100's of mg/day of gluten[1] where <50 mg/day is considered safe[2]

1) J A Syage, C.P Kelly, M A Dickason, A Cebolla-Ramizez, F Leon, R Dominguez, J A Sealey-Voyksner, Determination of Gluten Consumption in Celiac Disease Patients on a Gluten-Free Diet. Am. J. Clin. Nutr. 2018;107:201-207.
2) Catassi C, Fabianai E, Iacono G, et al. A prospective, double-blind, placebo-controlled trial to establish a safe gluten threshold for patients with celiac disease. Am J Clin Nutr 2007;85:160–6.

Dual-Enzyme Breakthrough Treatment

Latiglutenase is a clinically proven, non-absorbed, orally-administered biologic enzyme treatment that degrades gluten within the stomach



- Two recombinant gluten-specific enzymes work together to break down gluten in stomach
- Gluten fragments made too small to trigger the autoimmune response
- Safe, non-systemic drug

Competitive Landscape

Company	Asset	Phase	Efficacy	Administration	CeD Clinical Studies Completed
ZymagenX™	IMGX003 Latiglutenase	2b (Completed)	Mucosal protection and Strong symptom benefit	Oral	8
zedira® Dr. Falk	ZED1227	2a (Completed) 2b (Planned)	Histologic and symptomatic attenuation to gluten challenge	Oral	2
COUR Takeda	TAK-101 (CNP-101)	2a (Planned)	Reduction in IFN-γ Reduction in mucosal degradation (Phase I)	Infusion (8 days apart)	1
TOPAS THERAPEUTICS	TPM502 (Tolerase)	2a (Completed)	Reduction in IFN-γ and IL-2 relative to PBO for DQ-2.5 patients	Infusion (Days 1, 7, 15)	1
sanofi	Amlitelimab (OX40L mAb)	2a (Ongoing)	Reduction in histologic and symptom indicators	Subcutaneous Injection (approx monthly)	0
teva	TEV-53408	2a (Ongoing)	Reduction in villous atrophy (Vh:Cd)	Subcutaneous injection	0

Benefits of Latiglutenase vs. other candidates

- Three Phase 2 trials showing meaningful efficacy and safety including the only RCT demonstration
- Non-systemic, non-absorbed, inherently safe
- Oral administration
- Mechanism of action breaks down gluten toxin in the stomach avoiding an immunogenic event

Note: Includes companies entering or currently in at least Phase 2 development

Executive Leadership Combined With World Class Advisors and SAB

Jack Syage, PhD
CEO, Co-Founder

SAFRAN SYAGEN Caltech

Gail Comer, MD
Chief Medical Officer

Pfizer Wyeth Abbott

Jennifer Sealey Voyksner, PhD
CSO, Co-founder

LCMS Limited Boehringer Ingelheim The University of North Carolina at Chapel Hill

David Crean, PhD
Chief Business Officer

MEDICINOVA AQUA Pharmaceuticals an Almirall company Allergan

Chaitan Khosla, PhD
Special Advisor

ALVINE Pharmaceuticals, Inc. Stanford University

Steven E. Linberg, PhD
Clinical Operations Advisor

AIRWAY Therapeutics Chiesi

Phillip Lavin, PhD
Biostatistics Advisor

Harvard Medical School

Andrew E. Mulberg, MD
Regulatory Affairs Advisor

FDA

Lawrence Goldkind, MD
Regulatory Affairs Advisor

FDA

Joseph A. Murray, MD
Scientific Advisory Board

Mayo Clinic

Markku Mäki, MD, PhD
Scientific Advisory Board

Peter H.R. Green, MD
Scientific Advisory Board

Columbia University

Janice Soreth, MD
Regulatory & Clinical Operations

Scientific Advisory Board

FDA

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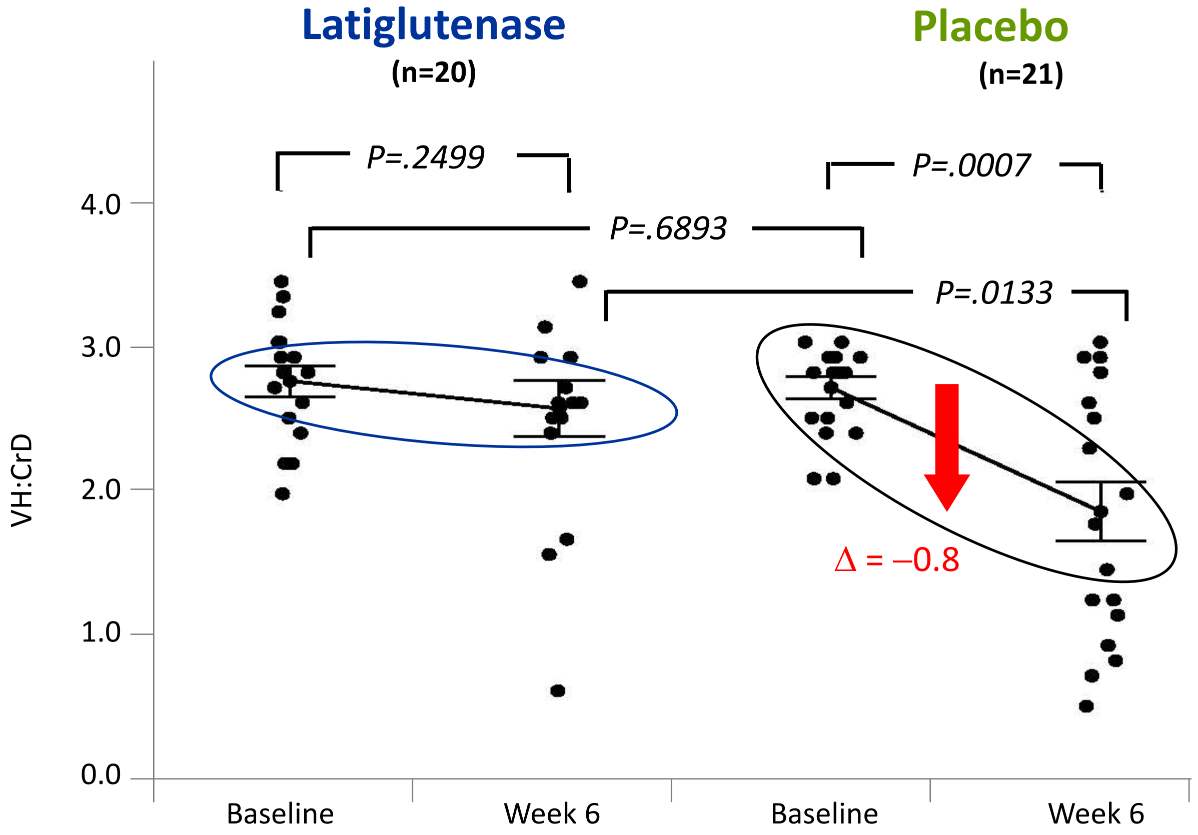
Section II

Phase 2 Results

Excellent Early Phase 2A Results

ALV003-1021

Phase 2A: Latiglutenase Preserves Intestinal Mucosa (VH:Cd)[1,2]



Gluten Challenge

- 2 g daily for 6 weeks
- Primary endpoints for Vh:Cd[2] and IEL[3]
- **Clinically and statistically significant protection of the mucosa ($p < 0.05$)**

1) Lahdeaho M-L, Kaukinen K, Laurila K, …Adelman DC, Maki M. Glutenase ALV003 Attenuates Gluten-Induced Mucosal Injury in Patients With Celiac Disease. Gastroenterology 2014;146:1649-1658
2) Vh:CD = Villious Height: Crypt Depth Ratio
3) IEL = Intreaepithelial Lymphocytes

Alvine/Abbvie Phase 2B Strategy

ALV003-1221

Primary Endpoint: Healing of the mucosa

- ***Premise:*** Latiglutenase protects the mucosa to *intended* gluten ingestion (Phase 2a); it should then protect the mucosa to *unintended* gluten ingestion in a real-world trial
- ***Outcome:*** Primary endpoint not met due to flawed clinical design; secondary endpoints (symptoms) showed clinically significant improvement in seroactive patients receiving ≥600 mg latiglutenase

Secondary Endpoint: Symptoms

CHANGE IN SYMPTOM SEVERITY FOR SEROACTIVE CELIAC PATIENTS OVER 12 WEEKS RELATIVE TO BASELINE

Symptom	Placebo	Latiglutenase (600 or 900 mg)	P-Value
	(N=54)	(N=49)	
Abdominal pain	-3.83	-8.79	0.008
Bloating	-4.86	-12.5	0.007
Tiredness	-5.84	-13.42	0.009
GI non-stool	-11.35	-24.37	0.009

Studies

CELIACTION STUDY

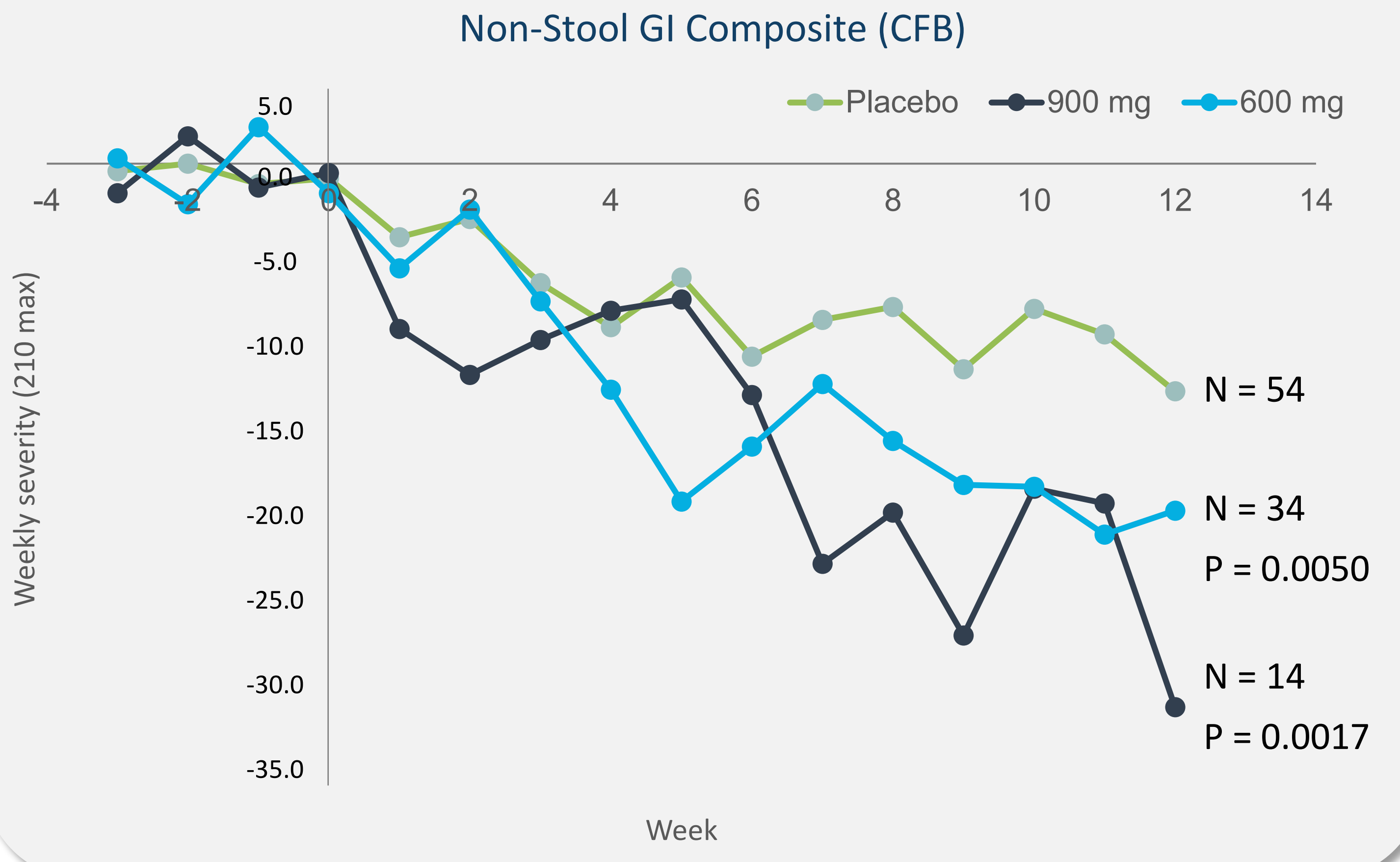
Murray JA, et al., No Difference Between Latiglutenase and Placebo in Reducing Villous Atrophy or Improving Symptoms in Patients with Symptomatic Celiac Disease. Gastroenterology 2017;152:787-798.

Syage JA, et al. Latiglutenase improves symptoms in seropositive celiac disease patients while on a gluten-free diet. Dig Dis Sci 2017;62:2428-2432.

Syage JA, et al. Latiglutenase Treatment for Celiac Disease: Symptom and Quality of Life Improvement for Seropositive Patients on a Gluten-Free Diet. GastroHep 2019;1:293-301.

Durable Symptom Benefit as Measured by Intervals

ALV003-1221



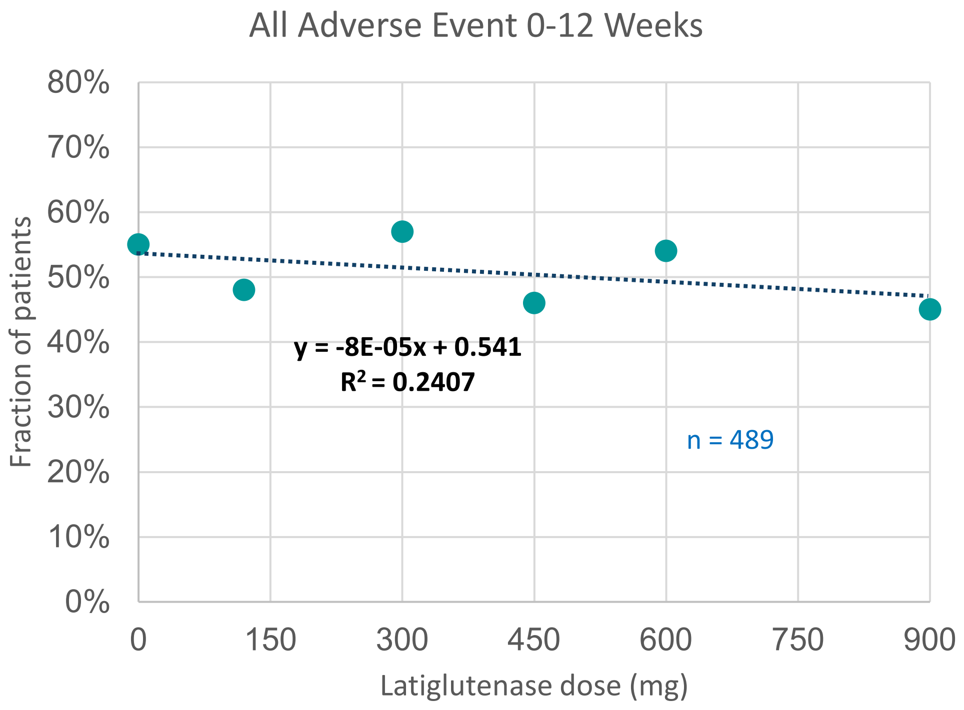
P value (unpaired t-test) for dose-dependent symptom improvement relative to placebo

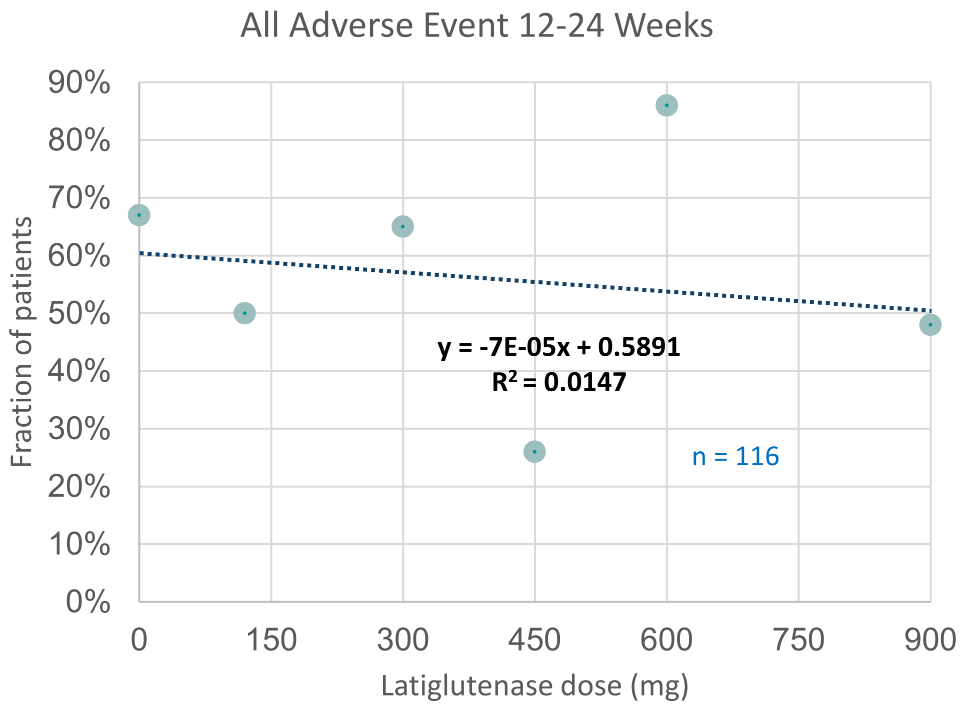
	600 mg	900 mg
Non-Stool GI Comp	0.005	0.002
Non-Stool Comp	0.008	0.006
Abdominal Pain	0.002	0.006
Bloating	0.014	0.006
Nausea	0.016	0.002
Tiredness	0.020	0.310

Excellent Safety Profile

ALV003-1221

All Adverse Events





No dose dependence on adverse events

Mayo Clinic Phase 2B Trial (CeliacShield) Summary

IMGX003-NCCIH-1721

Confirmation of effectiveness in this gluten-challenge study attests to latiglutenase (IMGX003) being leading candidate for Phase 3 success

Primary endpoint - histology

- ΔVh:Cd showed **88% attenuation** relative to placebo **(p = 0.057)**
- Other histologic measures had significance ranging from **p = 0.007 to 0.035**

Secondary endpoints

- Symptoms (abdominal pain, bloating, tiredness) **reduction ranging from 53% to 99%** relative to placebo and 3x2 week trend analysis significance ranging from **p = 0.002 to 0.030**
- Gluten in urine shows **95% reduction** relative to placebo with **p = 0.0008**
- Serology for all titer readings showed degradation for placebo and **improvement for latiglutenase**

Histology

IMGX003-NCCIH-1721

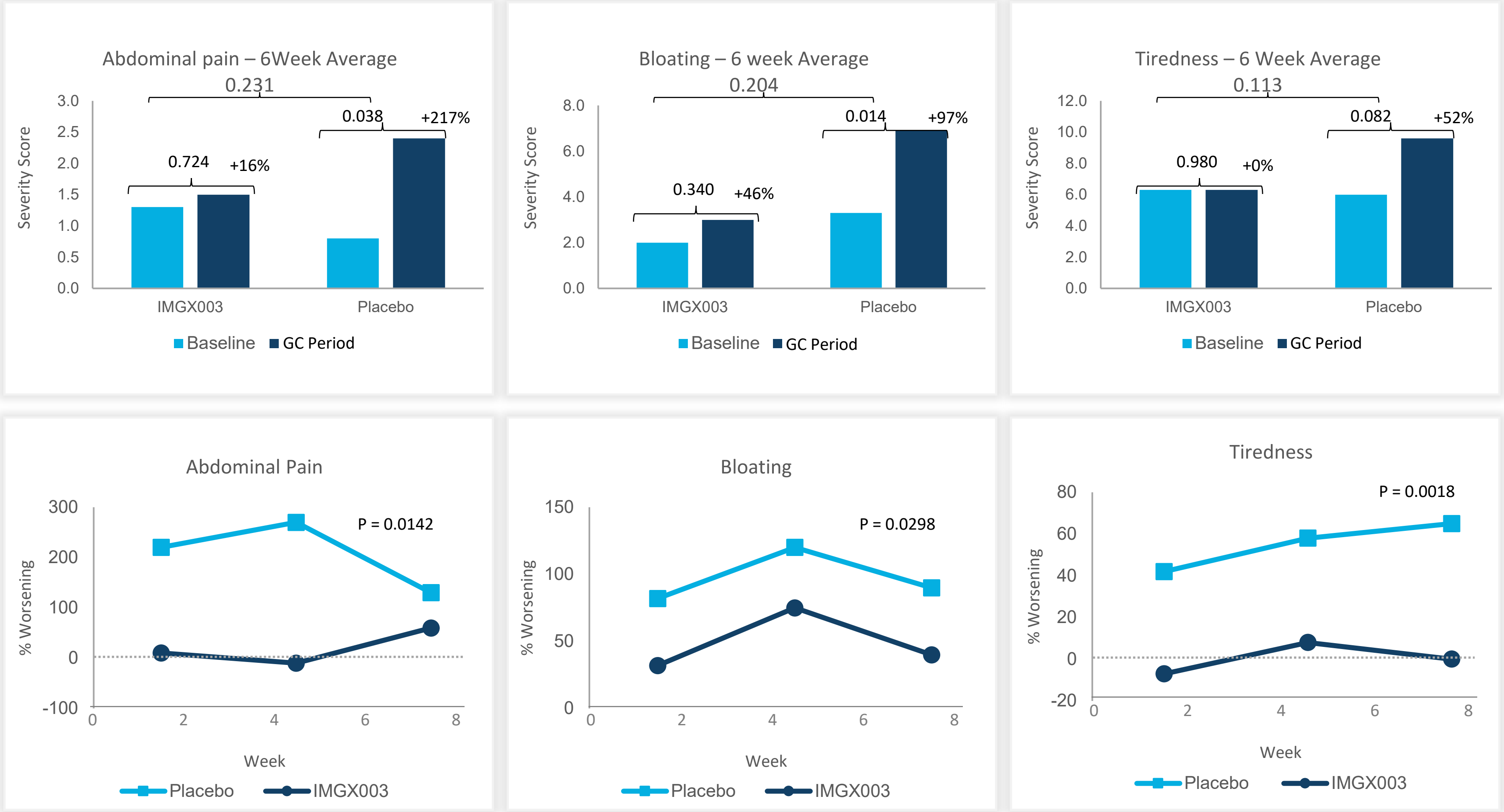
A = latiglutenase
P = placebo

Endpoints	Population	P-value	Reduction
Prospective Endpoints			Mean, median
Vh:Cd (mITT) – primary	n=21 (A), n=22 (P)	0.057	88%, 83%
IEL (mITT) – secondary	n=21 (A), n=22 (P)	0.018	60%, 72%
Retrospective Analysis			
Marsh-Oberhuber	n=21 (A), n=22 (P)	0.035	75%, 100%
VCIEL[1]	n=21 (A), n=22 (P)	0.007	70%, 76%

Note: A = Latiglutenase; P = placebo
1) A histology score that combines Vh:Cd and IEL measurements into a composite score VCIEL

Symptoms

IMGX003-NCCIH-1721



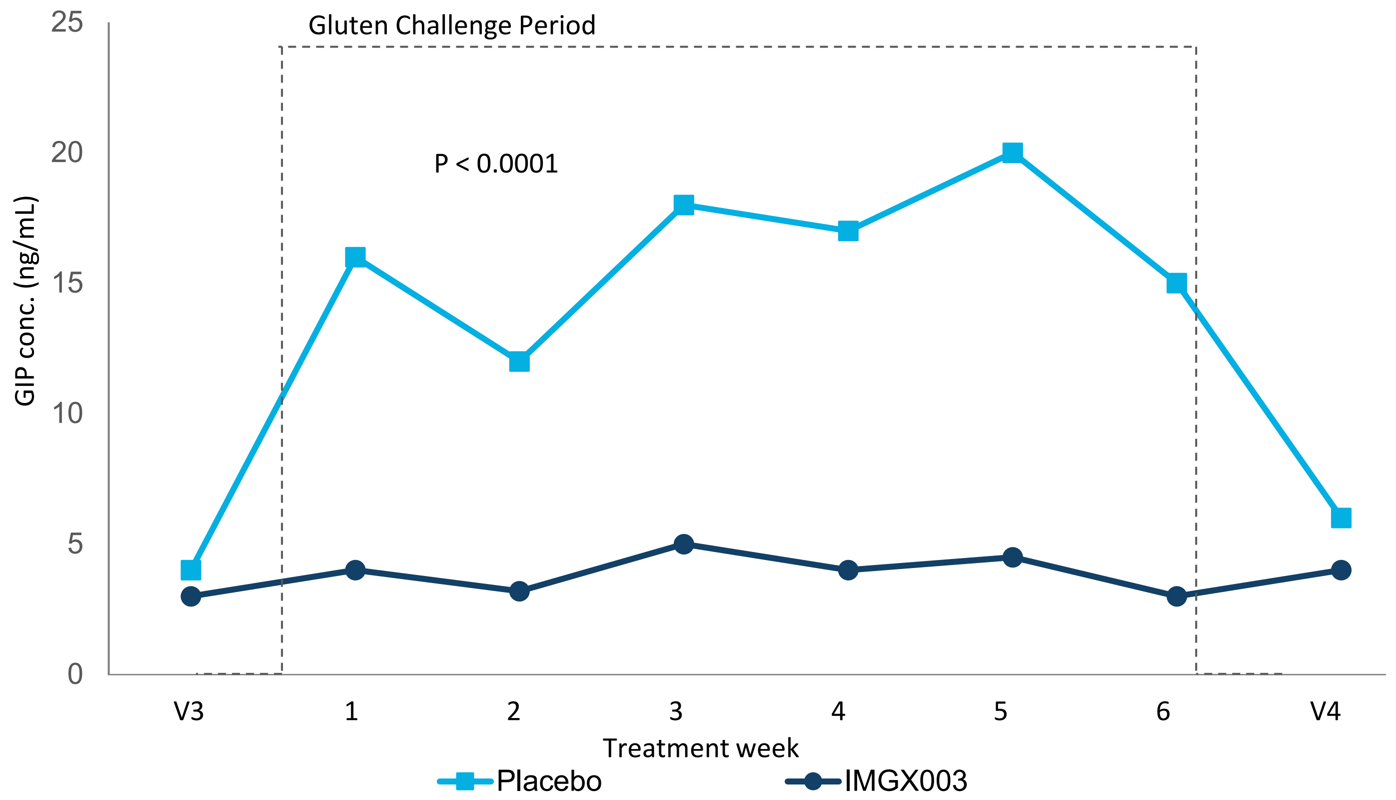
Latiglutenase attenuated symptoms for 2 g gluten per day by 93% (abdominal pain), 53% (bloating), 99% (tiredness). ANOVA p-values, respectively: 0.001, 0.006, 0.133

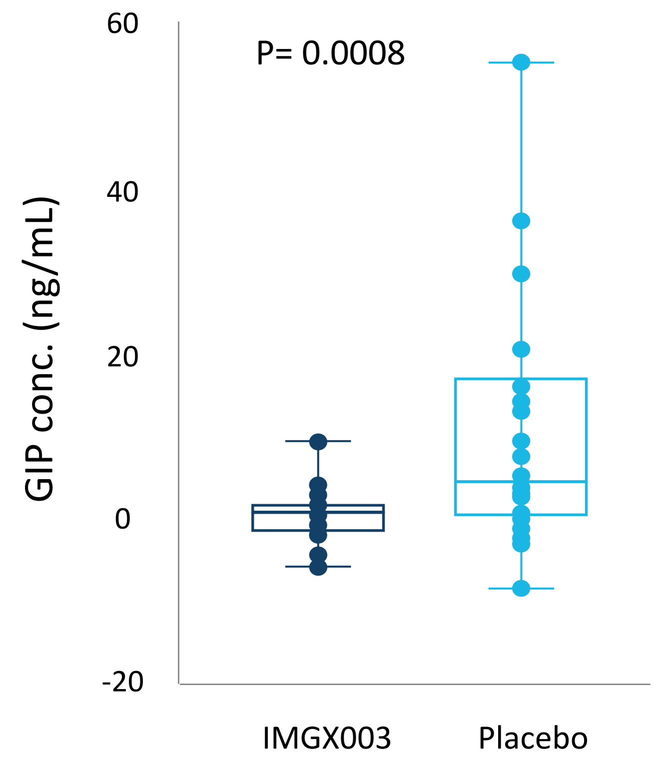
Statistically significant p-value for 2-week interval trend analysis for each symptom [unpaired, 2-tailed t-test]

IMGX003 (n=21)
Placebo (n=22)

Gluten Content in Urine

IMGX003-NCCIH-1721





- Latiglutenase attenuates gluten by approx. 95% based on ratio of means for deterioration for active vs. placebo
- Exceptional statistical significance for this direct measure of latiglutenase mechanism of action

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Section III

Phase 3 Plan

Phase 3 Protocol Aligned with FDA Input and Guidance

- FDA in accord with proposed Phase 3 endpoints, I/E criteria, outcome measuring tools & methods for Phase 3 study (Type C Meeting Aug 2022)

- End-of-Phase 2 Meeting with FDA evidenced further accord with non-clinical studies, endpoints, CTO and overall Phase 3 readiness (Type B Meeting April 2023)

- Type D meeting response (September 2024): single active arm is acceptable in the Phase 3 pivotal trial provided dose-ranging has been addressed. A gluten challenge study could be considered for this purpose

- Initial Pediatric Study Plan accepted by FDA (April 2024)

Phase 3 Overview

Label: FDA agreed with symptom reduction and histologic non-worsening in active symptomatic Celiac Disease patients

Ph 3-1: 26-week Phase 3 efficacy trial + 26-week safety run-out in symptom responders

Ph 3-2: 26-week gluten challenge + 26-week open label run-out

Efficacy endpoints (change from baseline to week 26)

- **Primary: Non-stool GI (abdominal pain, bloating, nausea) composite score reduction**
- **Key Secondary: Change from baseline improvement or non-worsening in histologic markers (Vh:Cd, IEL, VCIEL, Q-MARSH)**

Doses: 1,200 mg and placebo for efficacy and 600, 1,200 mg and placebo for gluten challenge

Stratify to:

- **Seroactive, Vh:Cd < 2.0**
- **Seroactive, 2.0 ≤ Vh:Cd ≤ 2.8**
- **Non-seroactive, Vh:Cd < 2.0 (optional)**
- **Seronegative, Vh:Cd > 2.0 (Gluten challenge)**

Total anticipated completed patients N = 600-660, which includes safety population of 200-220 completed dosed patients (to meet ICH guideline requirements).

Phase 3 Plan – Clinical Support of Primary & Secondary Endpoint

Primary Endpoint

Label: FDA agreed with symptom reduction and histologic non-worsening in active symptomatic CeD patients

Ph 3-1: 26-week Phase 3 efficacy trial + 26-week safety run-out in symptom responders

Ph 3-2: 26-week gluten challenge + 26-week open label run-out

Efficacy endpoints (change from baseline to week 26)

- **Primary: Non-stool GI (abdominal pain, bloating, nausea) composite score reduction**
- **Key Secondary: Change from baseline non-worsening in histologic markers (Vh:Cd, IEL, VCIEL)**

Doses: 1,200 mg and placebo for efficacy and 600, 1,200 mg and placebo for gluten challenge

Stratify to:

- **Seroactive, Vh:Cd < 2.0**
- **Seroactive, 2.0 ≤ Vh:Cd ≤ 2.8**
- **Seronegative, Vh:Cd > 2.0 (Gluten challenge)**

Total anticipated completed patients N = 600-660, which includes safety population of 200-220 completed dosed patients

Phase 2 Results

Alvine Phase 2B (Real world)

- Non-stool GI composite score [abdominal pain, bloating, nausea] change from baseline to week 12; **p < 0.01** in seroactive patients
- MMRM[1] for multiple symptoms **p < 0.01** (post hoc)

IMGX-Mayo Phase 2B (Gluten challenge)

- Non-stool GI composite score MMRM change from baseline to week 6; **p = 0.001 to < 0.05.**

Alvine Phase 2A (Gluten challenge)

- Abdominal pain **p = 0.047**, General GI Composite **p = 0.044** and QoL **p = 0.035**

MMRM: Mixed Model for Repeated Measures

Secondary Endpoint

Label: FDA agreed with symptom reduction and histologic non-worsening in active symptomatic CeD

Ph 3-1: 26-week Phase 3 efficacy trial + 26-week safety run-out

Ph 3-2: 26-week gluten challenge + 26-week open label run-out

Efficacy endpoints (change from baseline to week 26)

- **Primary: Non-stool GI (abdominal pain, bloating, nausea) composite score reduction**
- **Key Secondary: Change from baseline non-worsening in histologic markers (Vh:Cd, IEL, VCIEL)**

Doses: 1,200 mg and placebo for efficacy and 600, 1,200 mg and placebo for gluten challenge

Stratify to:

- **Seroactive, Vh:Cd < 2.0**
- **Seroactive, 2.0 ≤ Vh:Cd ≤ 2.8**
- **Seronegative, Vh:Cd > 2.0 (Gluten challenge)**

Total anticipated completed patients N = 600-660, which includes safety population of 200-220 completed dosed patients

Alvine Phase 2B (Real world)

- Vh:Cd non-worsening was evident but improvement (healing) based on change from baseline to week 12 not met

IMGX-Mayo Phase 2B (Gluten challenge)

- Vh:Cd worsening **attenuated by 80%** relative to placebo; **p = 0.057** [one outlier patient];
- Intra-epithelial lymphocytes (IEL); **p = 0.018;** VCIEL (Vh:Cd + IEL); **p = 0.007** (post hoc)

Alvine Phase 2A (Gluten challenge)

- Vh:Cd protection change from baseline; **p = 0.013**
- IEL protection to gluten-challenge based on change from baseline to week 6; **p < 0.015**

1) MMRM = Mixed-Effects Model for Repeated Measures

Section IV

Business Approach

Peer Review Support for Latiglutenase

Publications

Latiglutenase Improves Symptoms in Seropositive Celiac Disease Patients While on a Gluten-Free Diet
J. A. Syage, J. A. Murray, P. H. R. Green, C. Khosla. Dig. Dis. & Sci., 62, 2428-32 (2017)

Determination of Gluten Consumption in Celiac Disease Patients on a Gluten-Free Diet
J. A. Syage, C. P. Kelly, M. A. Dickason, A. Cebolla-Ramirez, F. Leon, R. Dominguez, J. A. Sealey-Voyksner. Am. J. Clin. Nutr., 107, 201-207 (2018)

Latiglutenase Treatment for Celiac Disease: Symptom and Quality of Life Improvement for Seropositive Patients on a Gluten-Free Diet
J. A. Syage, J. A. Murray, P. H. R. Green, C. Khosla, D. C. Adelman, J. A. Sealey-Voyksner. GastroHep, 1, 293-301 (2019)

Modeling Symptom Severity and Estimated Gluten Ingestion in Celiac Disease Patients on Gluten-Free Diet
J. A. Syage, P. T. Lavin. Int. J. Celiac Dis., 8, 85-89 (2020)

Latiglutenase Protects the Mucosa and Attenuates Symptom Severity in the CeliacShield™ Gluten Challenge Study
J. A. Murray, J. A. Syage, et al. Gastroenterology, 163, 1510-1521 (2022)

Dynamics of Serologic Change to Gluten in Celiac Disease Patients
J. Syage, A. Ramos, V. Loskutov, et al. Nutrients, 15, 5083 (2023)

A Composite Morphometric Duodenal Biopsy Mucosal Scale for Celiac Disease Encompassing both Morphology and Inflammation
J. A. Syage, M. Maki, D. A. Leffler, et al. Clin Gastro & Hep (2024)

NIH Grants

Latiglutenase Clinical Trials ($7.7M from NIH)

NIH grant from National Center for Complementary and Integrative Health (NCCIH) - $1.5M, 3-year grant

NIH grant from National Institute of Allergy and Infectious Diseases (NIAID) - $3.9M, 4-year grant

NIH grant from National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) - $2.3M, 3-year grant

ZymagenX in the News

Regular invited presentations at DDW, International CeD Symposium, Columbia CeD symposium, CDF symposium, Tampere International CeD Conference, etc.

Patent and Regulatory Protection

Robust patent portfolio (6 patent families; 4 for Latiglutenase)

- U.S.: 8 issued
- Foreign: 24 issued
- Patent protection (with 5-year extension for regulatory delay) until 2039
- New IP on CMC process in preparation

Expiry	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039

Generic Enzyme Degradation of Gluten

Enzyme – Combinations, Ratios

Formulations – Dosage, Excipients, Stabilizers

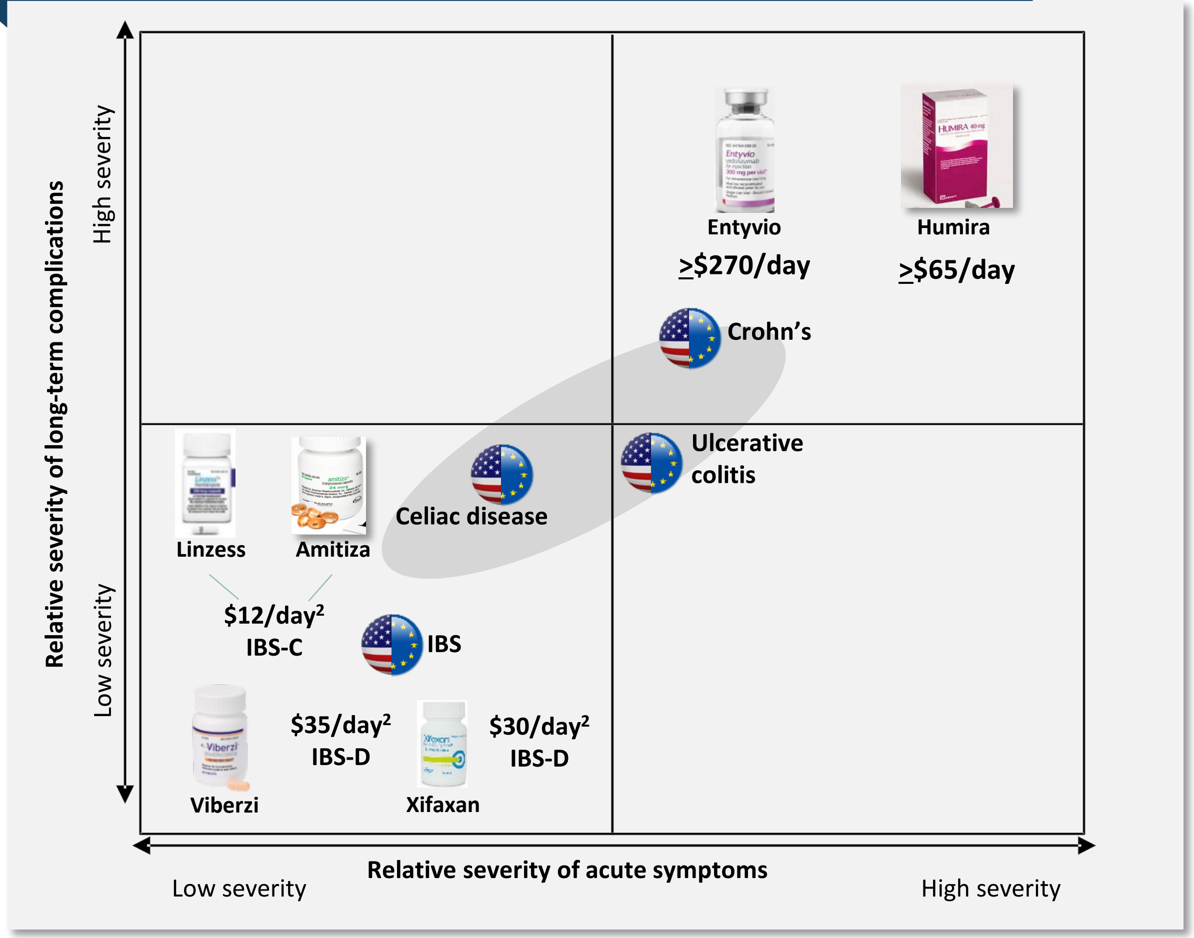
Patent Term Extension

Further regulatory exclusivities in U.S. and Europe for 10 years post-launch

- Currently anticipated to last until 2039 based on BLA in 2029

Significant Cost Burden & Payment Analogs

Comparable diseases and pricing analogs[1]



Drugs treating GI diseases with health impacts similar to CeD are analogs for Latiglutenase pricing

KOLs/Payors perceive CeD severity to be comparable to Ulcerative Colitis and Crohn's

Amitiza and Linzess (IBS-C, $12/day[2]) and Humira (Crohn's, >$65/day) cited as relevant comparators

Significant cost burden[3]

- Controls: $4.8K
- All CeD: $12.2K
- CeD partial remission: $18.2K

Note: All U.S. prices
1) Simon Kucher & Partners research completed in 2013
2) Prices represent current Wholesale Acquisition Cost
3) S. Guandalini, N. Tundia, R. Thakkar, D. Macaulay, K. Essenmacher, M. Fuldcore, "Direct cost in patients with celiac disease in the U.S.A.: A retropspective claims analysis," Dig Dis Sci 2016;10:2823-2830.

Reimbursement

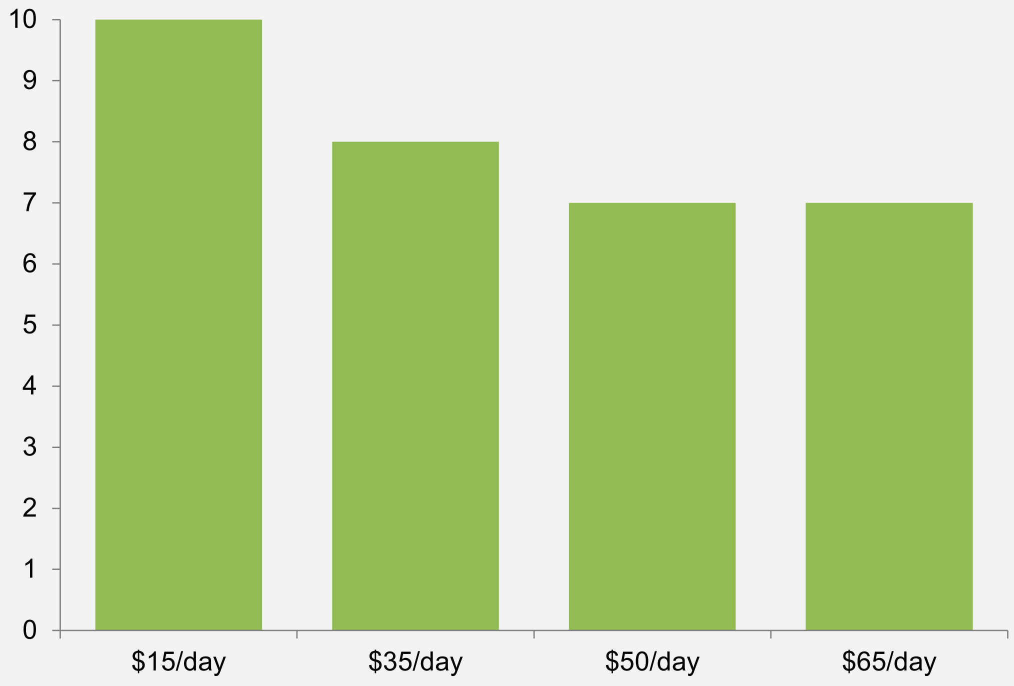
SURVEY PAYOR OVERVIEW

Xcenda
AmerisourceBergen

- Xcenda, an AmerisourceBergen company performed payor study (2017)
- 10 payors surveyed covering ~64M U.S. lives
- Surveyed payors strategically targeted, by size, type and geographic coverage

CONCLUSION

- All surveyed payors supported $15 to >$65/day range with 70% supporting >$65/day
- Payor supported pricing is highly favorable and in-line with other GI diseases



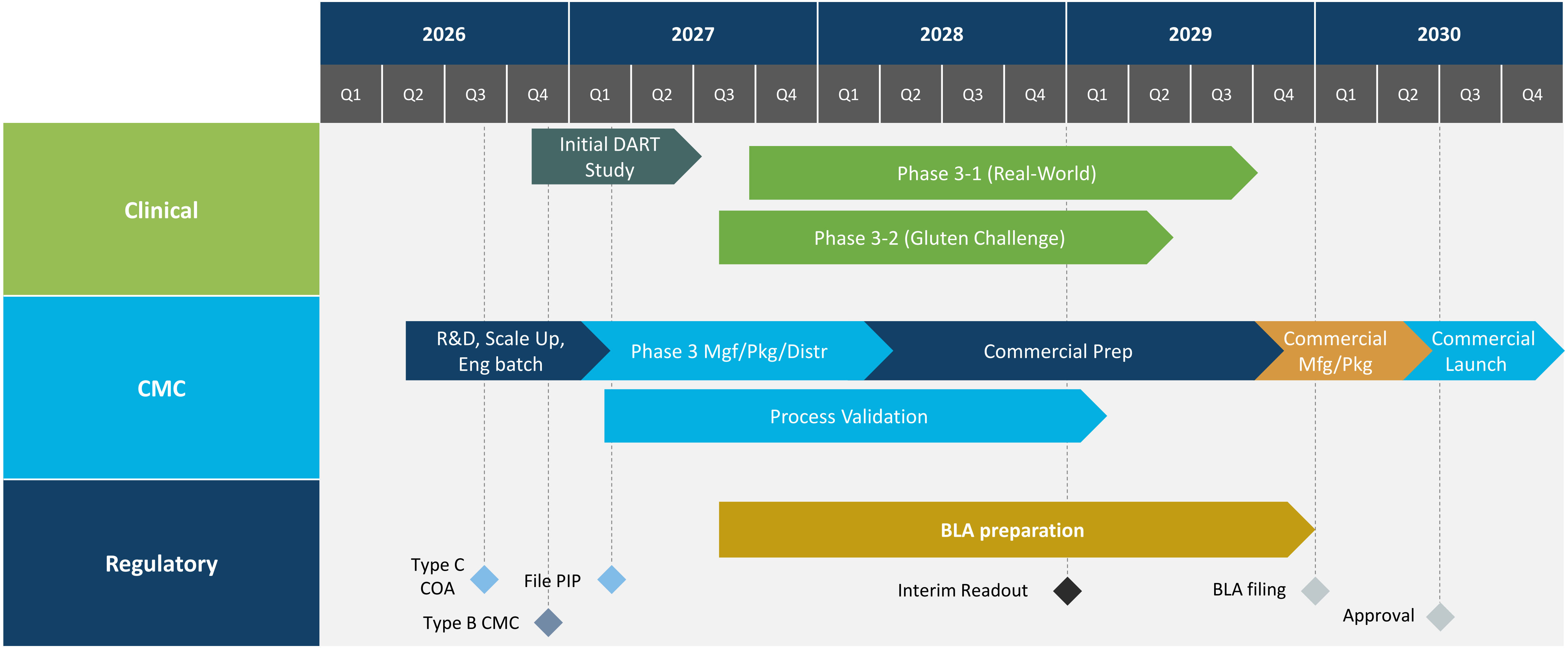
- Reimbursement code based on a prescription
- Outsourced manufacturing
- Distribution via strategic partner

Chemistry Manufacturing & Controls

CMOs for Biological Drug Substance (BDS) and Finished Drug Product (FDP) manufacturing, primary packaging and clinical packaging and distribution

- Manufacture of Ph3 and commercial BDS (ACS Dobfar)
- FDP formulation/primary packaging (Catalent, Ropack)
- Flavor manufacturing and packaging (Corealis/ Ropak)
- Analytical, characterization and stability testing (ACS Dobfar, Alcami, Charles River)
- Clinical packaging and distribution (PCI)

Timeline to Market



Section V

Investor Opportunity

Funding Opportunities

Exit Plan

Citizens Bank is representing ZymagenX in M&A and partnering discussions

Convertible Note

$2-4M Offering

10% div, 24% disc, $27M cap value

Recent Comparable Pre-Revenue Valuations

Target	Acquirer	Description	Cash Price
Provention Biosystems	Sanofi (Mar 2023)	Pre-commercial FDA-approved T1D delayed onset drug and a Phase 2b CeD drug	$2.9B
Kymab	Sanofi (Jan 2021)	KY1005 monoclonal antibody for immune-mediated, immuno-oncology and inflammatory disorders	$1.1B
Emisphere	Novo Nordisk (Dec 2020)	GLP-1 candidate for Type 2 diabetes and Eligen drug delivery technology	$1.35B
Forty Seven	Gilead (Mar 2020)	Magrolimab investigational monoclonal antibody candidate for immuno-oncology	$4.9B
Dermira	Eli Lilly (Feb 2020)	Lebrikizumab in Phase 3 for treatment of moderate-to-severe atopic dermatitis	$1.1B
Synthorx	Sanofi (Jan 2020)	THOR-707 preclinical immuno-oncology therapy and lymphokines discovery platform	$2.5B

Highlights

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3. End-of-Phase 2 Meeting completed and Phase 3 trial in preparation
4. Highly experienced and successful management team and scientific advisory board
5. Significant value creation and early exit potential